WS-03: Test Card TC-03 — Moving Vehicle Advance-Ratio Sweep

AERODYNAMIC POLAR MAPPING, FORWARD THRUST & LIFT (TRUCK TEST PYLON)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** TC-03	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Aerodynamic Force & Polar Sweep vs Forward Airspeed

Mount aircraft to 6-axis load cell pylon above open-bed truck. Drive runway in steady speed steps in calm air (< 1.0 m/s wind).

Run	Truck Speed (V_{inf})	Flap Freq	Thrust ($3F_{xS}$) Target	Measured $3F_{xS}$	Target Lift ($3L_S$)	Measured Lift	Pitch Moment ($3M_{yS}$)	Pitot Speed
R1	4.0 m/s (9 mph)	2.20 Hz	*≥ 12.0 N*	[ ] N	*≥ 20.0 N*	[ ] N	[ ] N·m	[ ] m/s
R2	6.2 m/s (14 mph)	2.30 Hz	*≥ 15.0 N*	[ ] N	*≥ 56.0 N (Unstick)*	[ ] N	[ ] N·m	[ ] m/s
R3	9.0 m/s (20 mph)	2.40 Hz	*≥ 8.0 N*	[ ] N	*≥ 56.0 N (Cruise)*	[ ] N	[ ] N·m	[ ] m/s
R4	12.0 m/s (27 mph)	2.60 Hz	*≥ 5.0 N*	[ ] N	*≥ 58.0 N*	[ ] N	[ ] N·m	[ ] m/s
R5	14.0 m/s (31 mph)	2.80 Hz	*≥ 2.0 N (Sprint)*	[ ] N	*≥ 62.0 N*	[ ] N	[ ] N·m	[ ] m/s

2. Pure Passive Glide Aerodynamics Verification (Wings Stationary: f = 0.0 Hz)

Hold wings completely stationary at clean cruise pitch (α = +4.0°). Measures baseline fixed-wing lift deficit vs airspeed.

Truck Speed (V_{inf})	Flapping State	Expected Passive Lift	Measured Passive Lift	Measured Drag ($3D_S$)	Measured $3L/3D_S$ Ratio	Flight Safety Note
9.0 m/s (20.1 mph)	**0.0 Hz (Fixed)**	*37.13 N (18.6 N deficit)*	[ ] N	[ ] N	[ ]	Cannot sustain 1g level flight
11.0 m/s (24.6 mph)	**0.0 Hz (Fixed)**	*55.77 N (1.00g balance)*	[ ] N	[ ] N	[ ]	**Safe unpowered dead-stick glide**
13.0 m/s (29.1 mph)	**0.0 Hz (Fixed)**	*77.80 N (excess lift)*	[ ] N	[ ] N	[ ]	Positive climb glide flare

3. Pitot Differential Pressure Sensor vs True Truck Ground Speed

Truck GPS Speed: 6.2 m/s	**Target Counts:** ~56 counts (23.5 Pa)	**MS4525DO Counts:** [ ]	**Indicated Airspeed:** [ ] m/s
Truck GPS Speed: 9.0 m/s	**Target Counts:** ~118 counts (49.6 Pa)	**MS4525DO Counts:** [ ]	**Indicated Airspeed:** [ ] m/s

GATE CERTIFICATION: TC-03 ADVANCE-RATIO TRUCK SWEEP QUALIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE:** _____
REMARKS / DISCREPANCIES: _____	